Swami Keshvanand Institute of Technology, Management & Gramothan,
Ramnagaria, Jagatpura, Jaipur-302017, INDIA
Approved by AICTE, Ministry of HRD, Government of India
Recognized by UGC under Section 2(f) of the UGC Act, 1956
Tel. : +91-0141- 5160400 Fax: +91-0141-2759555
E-mail: info@skit.ac.in Web: www.skit.ac.in

असतो मा सद्गमय

UNIT-5

# Graphs

Basic Concept, Different representation of Graph, Graph Traversal (BFS & DFS), Minimum Spanning Tree (Prims & kruskal) Dijkstra's shortest path algorithm.

Hashing: Hash function, Address calculation technique, common hashing functions

Collision resolution: Linear and Quadratic probing, Double Hashing

Swami Keshvanand Institute of Technology, Management & Gramothan,
Ramnagaria, Jagatpura, Jaipur-302017, INDIA
Approved by AICTE, Ministry of HRD, Government of India
Recognized by UGC under Section 2(f) of the UGC Act, 1956
Tel. : +91-0141- 5160400 Fax: +91-0141-2759555
E-mail: info@skit.ac.in Web: www.skit.ac.in

Graph -

A nonlinear data structure. A graph G consists of two things:

(1) A set V of element called nodes
(2) A set E of edges such that each edge e in E is identified with a unique pair [u,v] of nodes in V, denoted by e = [u,v]

$G = (V, E)$

Suppose $e = [u,v]$,

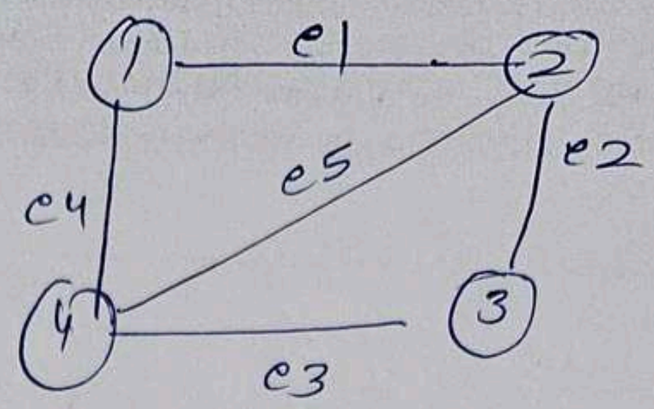
u,v are said to be adjacent nodes or neighbors

* If u does not belong to any edge then u is called an isolated node.

Basic Concept

* path - is a sequence of vertices & edges

If no. of nodes are - N
then no of edges are = N-1




$V = \{1, 2, 3, 4\}$

$e = \{e1, e2, e3, e4\}$ or

$\{(1,2)(2,3),(3,4)(4,1)\}$

* Simple path - If all the nodes are distinct with the exception that $V_0$ to $V_n$ (start and end)

* Cycle - is a closed simple path

* closed path - If a path have same initial and End vertex.

* Ex

1 — 2 — 3 — 4 → path

1 — 2 — 3 — 4 — 2 — 1 → closed path

1 — 2 — 3 — 4 — 1 — simple path

1 — 2 — 3 — 4 — 1 → cycle

5-cycle

no. of vertex

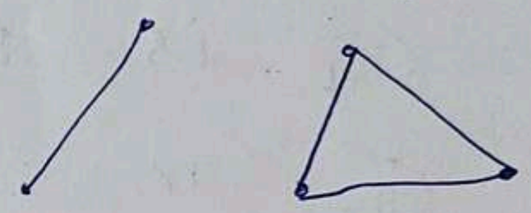
Swami Keshvanand Institute of Technology, Management & Gramothan,
Ramnagaria, Jagatpura, Jaipur-302017, INDIA
Approved by AICTE, Ministry of HRD, Government of India
Recognized by UGC under Section 2(f) of the UGC Act, 1956
Tel. : +91-0141- 5160400 Fax: +91-0141-2759555
E-mail: info@skit.ac.in Web: www.skit.ac.in

## Connected & Disconnected graph -

A graph G is said to be connected if there is a path blw any two of its nodes or A graph G is connected if and only if there is a simple ~~graph~~ path blw any two nodes in G.



Connected graph
$G_1$

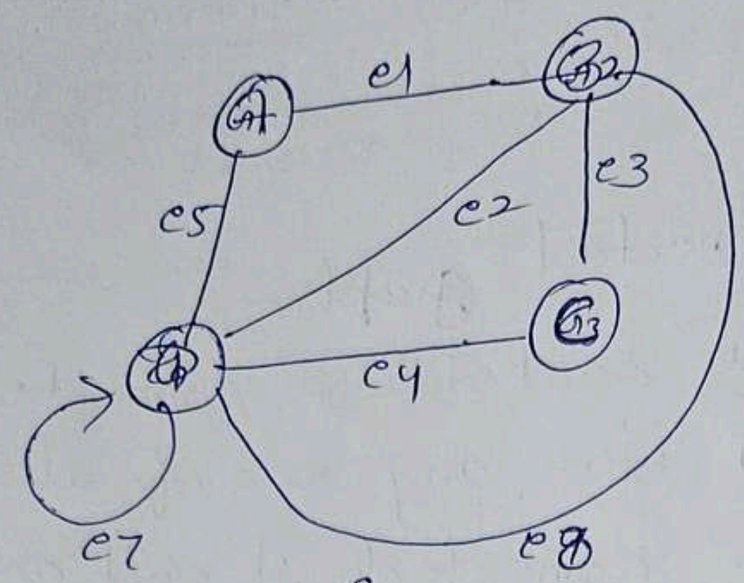


$G_2$
Disconnected graph

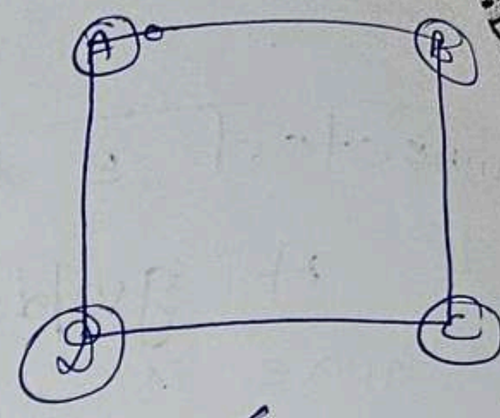
* A connected graph without any cycle is tree.

A Simple graph & multigraph -

A multigraph may contain either self loop or parallel edges in the graph




$G_1$




$G_2$

e2 and e8 are multiple edges or parallel edges

e7 e8 — is self loop.

So $G_1$ is multigraph

$G_2$ is Simple graph

* Directed Graph -

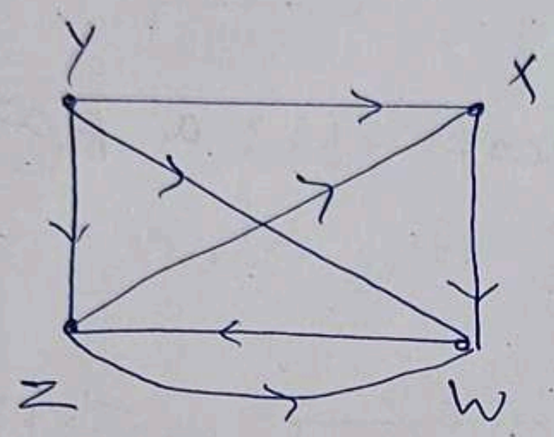
A directed graph or digraph, is the same as a multigraph except that each edge is assigned a direction rather then unordered pair [U,V].

A directed graph said to strongly connected if each pair (U,V) of nodes in G there is a path from U to V and there is path from V to U also.

Swami Keshvanand Institute of Technology, Management & Gramothan,
Ramnagaria, Jagatpura, Jaipur-302017, INDIA
Approved by AICTE, Ministry of HRD, Government of India
Recognized by UGC under Section 2(f) of the UGC Act, 1956
Tel. : +91-0141- 5160400 Fax: +91-0141-2759555
E-mail: info@skit.ac.in Web: www.skit.ac.in

# Sequential Matrix Repn

$$a_{ij} = \begin{cases} 1 & \text{if } v_i \text{ is adjacent to } v_j \\ 0 & \text{otherwise} \end{cases}$$




The adjacency matrix of a graph G does not depend on the ordering of the nodes

$$A = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix},$$

If G is an undirected graph, then the adjacency matrix A of G will be a symmetric matrix

Similarly $A^2, A^3 \ldots$ are the adjacency matrix of graph G, where 2 refer the path of length (2), as in graph

X − W − Z = path length is 2

W − Z − X = " "

for the given graph

$$A^2 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 2 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix} \quad A^3 = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 1 & 0 & 2 & 2 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} A^3 = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 2 & 0 & 2 & 3 \\ 1 & 0 & 1 & 2 \\ 1 & 0 & 1 & 1 \end{pmatrix}$$

(path of length 2) path of length 3 & 4 resp.

Now, the matrix $B_r = A + A^2 + A^3 + \cdots A^r$

Path matrix.

$$P_{ij} = \begin{cases} 1 & \text{if there is a path } v_i, v_j \\ 0 & \text{otherwise} \end{cases}$$

$B_m = A + A^2 + A^3 + \cdots A^m$

$$B_4 = \begin{pmatrix} 1 & 0 & 2 & 3 \\ 5 & 0 & 6 & 8 \\ 3 & 0 & 3 & 5 \\ 2 & 0 & 3 & 3 \end{pmatrix} \text{ and } P = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{pmatrix}$$

Means the node $v_2$ is not reachable from any of the other node.

* A graph is strongly connected if and only if the path matrix P of G has no zero entries.

Swami Keshvanand Institute of Technology, Management & Gramothan,
Ramnagaria, Jagatpura, Jaipur-302017, INDIA
Approved by AICTE, Ministry of HRD, Government of India
Recognized by UGC under Section 2(f) of the UGC Act, 1956
Tel. : +91-0141- 5160400 Fax: +91-0141-2759555
E-mail: info@skit.ac.in Web: www.skit.ac.in

## Linked Representation of a Graph -

The linked Repⁿ contain two list
1) a Node list 2) An edge list

Node list

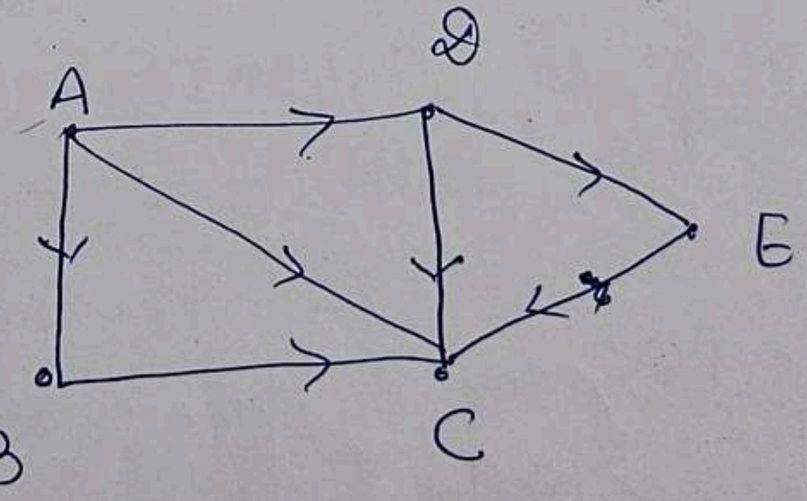
| NODE | NEXT | ADJ | |
|---|---|---|---|

Each element in the list will correspond to a node in G,

Edge list - | DEST | link | |

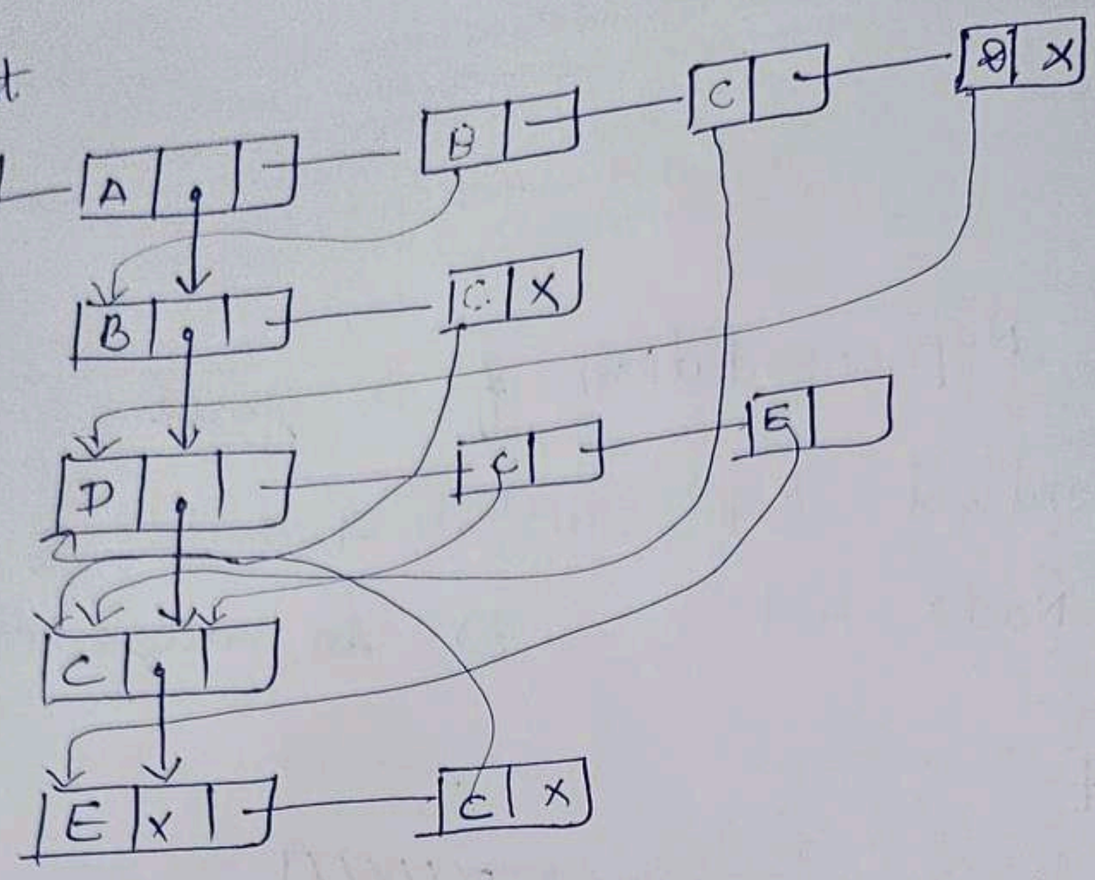
Each element in the list edge will correspond to an edge of G and will be a record of the form:

Graph




Adjacency list

A — B C D
B — C
C —
D — C, E
E — C




The repn may also include an array weight or may include an array edge when graph is a labeled graph.

Swami Keshvanand Institute of Technology, Management & Gramothan,
Ramnagaria, Jagatpura, Jaipur-302017, INDIA
Approved by AICTE, Ministry of HRD, Government of India
Recognized by UGC under Section 2(f) of the UGC Act, 1956
Tel. : +91-0141- 5160400 Fax: +91-0141-2759555
E-mail: info@skit.ac.in Web: www.skit.ac.in

## Warshall's Algorithm -

Let G be a directed graph with m nodes $V_1, V_2, \ldots V_m$. This algorithm find the shortest path in G, when G is weighted.

There is a simple path from $V_i$ to $V_j$ (either direct or through with other node)

$V_i \longrightarrow V_j$

$V_i \longrightarrow V_k \longrightarrow V_j$

[All source shortest path]

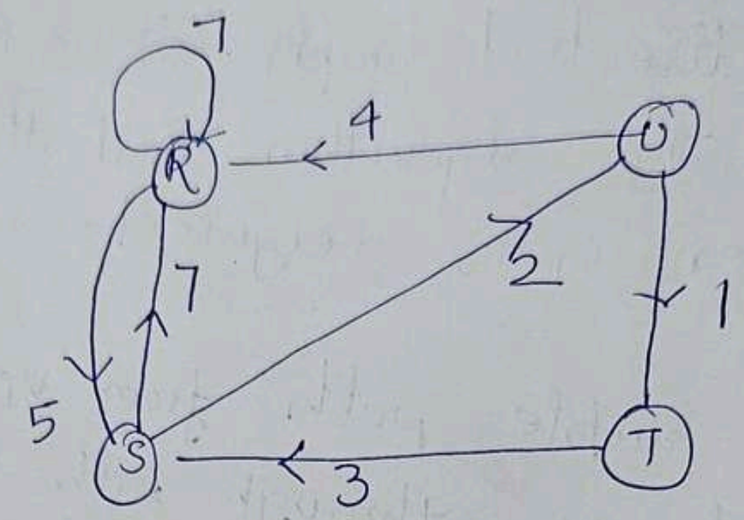
Warshall's Algorithm ( G, M (nodes))

1. Repeat for I, J = 1, 2 ... M: [Initializes P]
   If A[I, J] = 0 then set P[I, J] = 0
   Else set P[I, J] = 1
2. Repeat steps 3 and 4 for K = 1, 2, .. M:
3. Repeat step 4 for I = 1, 2, .. M:
4. Repeat for J = 1, 2 ... M:

setp[I, J] = P[I, J] ∨ ( P(I, k) ∧ P(

5. Exit.

Eg




$$W = \begin{array}{c} \\ R \\ S \\ T \\ O \end{array} \begin{array}{c} \begin{array}{cccc} R & S & T & O \end{array} \\ \begin{bmatrix} 7 & 5 & 0 & 0 \\ 7 & 0 & 0 & 2 \\ 0 & 3 & 0 & 0 \\ 4 & 0 & 1 & 0 \end{bmatrix} \end{array} = Q_0 = \begin{bmatrix} 7 & 5 & \infty & \infty \\ 7 & \infty & \infty & 2 \\ \infty & 3 & \infty & \infty \\ 4 & \infty & 1 & \infty \end{bmatrix}$$

$$Q_k[i,j] = MIN\Big(\big(Q_{k-1}[i,j]\big), Q_{k-1}[i,k] + Q_{k-1}[k,j]\Big)$$

$$Q_1 = \begin{bmatrix} 7 & 5 & \infty & \infty \\ 7 & \underline{12} & \infty & 2 \\ \infty & 3 & \infty & \infty \\ 4 & \underline{9} & 1 & \infty \end{bmatrix}$$

$$Q_1(S,S) = [\infty, Q(S,R) + Q(R,S)]$$

$$= 12$$

Swami Keshvanand Institute of Technology, Management & Gramothan,
Ramnagaria, Jagatpura, Jaipur-302017, INDIA
Approved by AICTE, Ministry of HRD, Government of India
Recognized by UGC under Section 2(f) of the UGC Act, 1956
Tel.: +91-0141-5160400 Fax: +91-0141-2759555
E-mail: info@skit.ac.in Web: www.skit.ac.in

## Traversing a Graph-

* Two standard ways first, this ~~Breadth~~ Traversing-
① Breadth first search ② Depth first search
During the execution of algorithm, each Node N of G will be in one of the states
- Ready - Status-1 - Initial node
- Waiting - Status-2 node in queue or stack
- Processed - Status-3 node has been processed

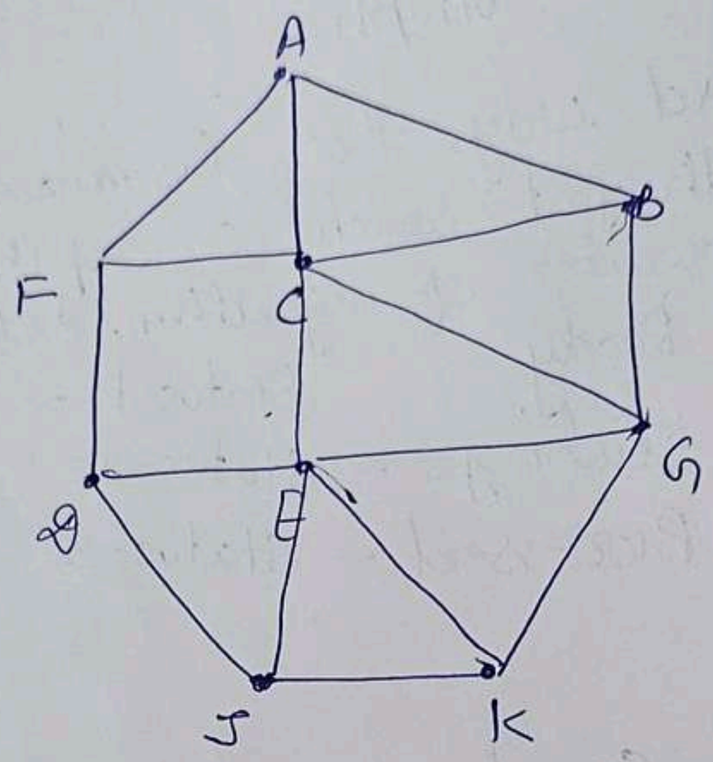
### Breadth-first Search -

This is accomplished by using a queue to hold nodes that are waiting to be processed.
The algorithm is as follows -

1. Initialize all nodes to the ready state (STATUS = 1)
2. Put the starting node A in Queue and change its status to the waiting state (status = 2)
3. Repeat step 4 & 5 until Queue is empty
4. Remove front node of Queue. Process it (Status = 3)
5. Add to the rear of Queue all

the neighbors of N, that are in ready st
and change their state (status = 2)

6 Exit




Adjaceney L

| | |
|---|---|
| A | F C B |
| B | G C |
| C | F |
| D | C |
| E | D C J |
| F | D |
| G | C E |
| J | D K |
| K | E G |

step1 Q: [A]

Status of A = 2
all other nodes status is = 1

step 2 Q: [B|C|F]

A,

status of A = 3
B, C, F = 2
other nodes = 1

step 3 Q: [C|F|G]

A, B,

Status of B = 3
C, F, G = 2
other node = 1

step 4 Q. [F|G]

A, B, C

Status of C = 3
F, G = 2

ASV PAPERS PVT. LTD.
3, Behind Hotel Vikrant,
Old Bus Stand, Beawar-305 901 (Raj.)
email @ : info@asvpapers.in
website : www.asvpapers.in

Swami Keshvanand Institute of Technology, Management & Gramothan,
Ramnagaria, Jagatpura, Jaipur-302017, INDIA
Approved by AICTE, Ministry of HRD, Government of India
Recognized by UGC under Section 2(f) of the UGC Act, 1956
Tel. : +91-0141- 5160400 Fax: +91-0141-2759555
E-mail: info@skit.ac.in Web: www.skit.ac.in

step 5 : Q: [ | G | Θ ]
A, B, C, F

Status of F = 3
Status of G, Θ = 2
other = 1

step 6 Q: [ | Θ | E ]
G

Status of G = 3
Θ, E = 2

step 7 Q: [ | E | ]
Θ

step 8 Q [ | J | ]
E

step 9 Q: [ | K ]
J

step 10 Q [ ]
K stob

step 11: Q [ ] → empty
stop

stop

A B C F G Θ E J K G

BPS

Depth first search

( Algorithm is similar to DFS, only change is stack rather than queue)

for the same graph

DFS is : A B G E J K D C F

Another DFS for the same graph is

J K G C F E D

( A, B not visited as they are not reachable by J)

step1 ① find out the differences b/w BFS & DFS

step 2

step 3 Complexity of BFS & DFS

Hist → O(V+E) bcoz every vertex and every edge explored only once.

step 4

A/L

BFS — Queue
DFS — Stack

Adjacency matrix — $O(V^2)$

SKIT
असतो मा सद्गमय
Swami Ke
Appro
Recogni
Tel
E-ma
POSETS -
Subha
Vi

Swami Keshvanand Institute of Technology, Management & Gramothan,
Ramnagaria, Jagatpura, Jaipur-302017, INDIA
Approved by AICTE, Ministry of HRD, Government of India
Recognized by UGC under Section 2(f) of the UGC Act, 1956
Tel. : +91-0141- 5160400 Fax: +91-0141-2759555
E-mail: info@skit.ac.in Web: www.skit.ac.in

असतो मा सद्गमय

# POSETS -

Suppose S is a graph such that each node $v_i$ of S represent a task and each edge (u,v) means that the completion of the task u is a prerequisite for starting the task v.

Suppose S is a graph without cycle.
Consider a Relation '<' on S as

$u < v$ [ if there is a path from u to v ]

The relation has the following three properties

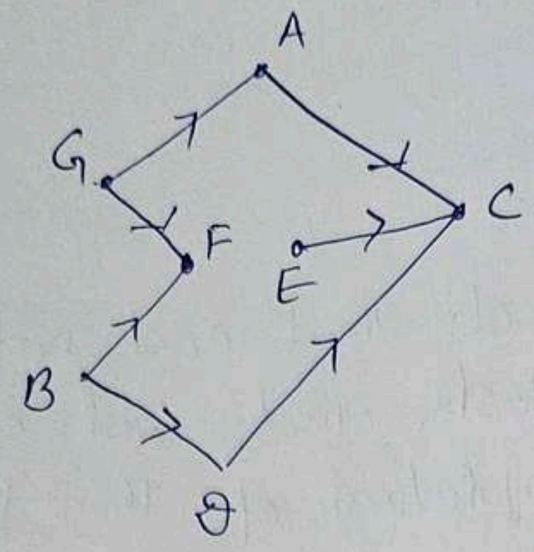
(1) For each element u in S, we have $u \not< u$
( Irreflexivity )

(2) If $u < v$ then $v \not< u$ ( Asymmetry )

(3) If $u < v$ And $v < w$ then $u < w$
( Transitivity )

This relation < on S is partial order of S and S is Partial ordered set or poset

Ex. Let S be the graph




(a)

Adjacency list

| | |
|---|---|
| A | C |
| B | ϑ F |
| C | |
| ϑ | C |
| E | C |
| F | |
| G | A, F |

(b)

## Topological sorting -

**Step** Topological sorting is possible in directed graph without cycle (or a poset). A topological

Step 2 sort T of S is a linear ordering of the node

of S which preserves the original poetial ordering

of S. That is if $U < V$ in S, then U comes

Step 3 before V in the linear ordering T.

Accordingly, our algorithm will repeat the

Step 4 following two step until the graph is empty

A,

(1) finding a Node N with zero indegree

(2) deleting N and its edges from the graph

Swami Keshvanand Institute of Technology, Management & Gramothan,
Ramnagaria, Jagatpura, Jaipur-302017, INDIA
Approved by AICTE, Ministry of HRD, Government of India
Recognized by UGC under Section 2(f) of the UGC Act, 1956
Tel. : +91-0141- 5160400 Fax: +91-0141-2759555
E-mail: info@skit.ac.in Web: www.skit.ac.in

Algo. Topo_Sort( S )

1. find the indegree Indeg (N) of each node N of S.
2. Put in a queue all the nodes with zero indegree
3. Repeat steps 4 and 5 until the queue is empty
4. Remove the front Node N of the queue
5. Repeat the following for each neighbor m of the Node N:
   (a) set indeg (m) = Indeg (m) - 1
   (b) If Indeg (m) = 0 then Add M to the rear of the queue
6. Exit

1

| Node | A | B | C | D | G | F | E |
|---|---|---|---|---|---|---|---|
| Indeg | 1 | 0 | 3 | 1 | 0 | 2 | 0 |

2. Initially add to the queue

Qu | B | E | G |

3. B | E | G | D | Indeg D = 0, Indeg F = 1

4. B E | G | D | Indeg C = 2

5 B E G | D | Indeg A = 0, Indeg F = 0

B E G D | A | F | Indeg C = 1

Step

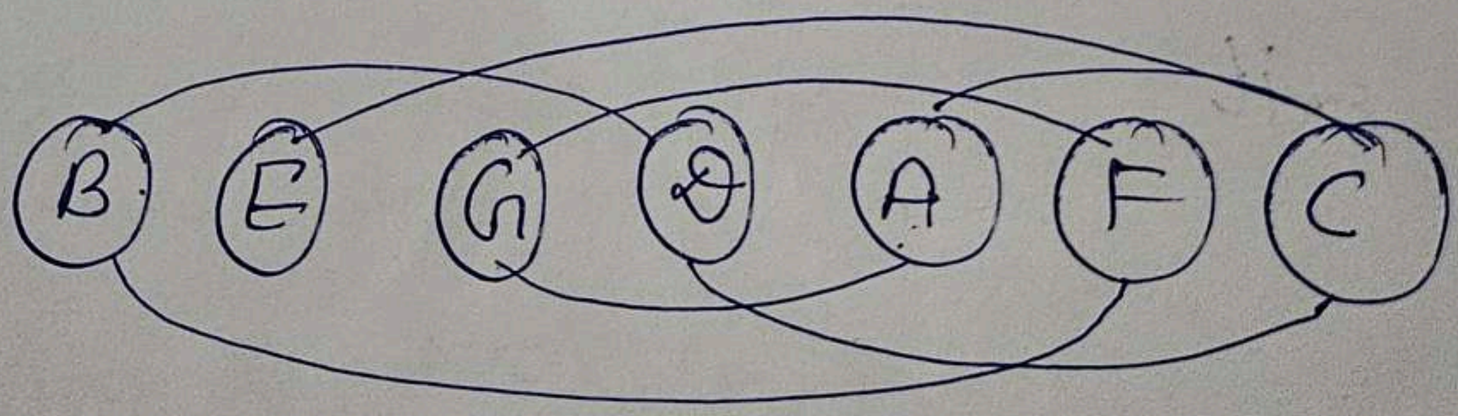
B E G D A | F | C | Indeg C = 0

step

B E G D A F

B E G D A F C

7:



Swami Keshvanand Institute of Technology, Management & Gramothan,
Ramnagaria, Jagatpura, Jaipur-302017, INDIA
Approved by AICTE, Ministry of HRD, Government of India
Recognized by UGC under Section 2(f) of the UGC Act, 1956
Tel. : +91-0141- 5160400 Fax: +91-0141-2759555
E-mail: info@skit.ac.in Web: www.skit.ac.in

# Shortest Path Algorithms

To calculate the shortest path btw the ~~two~~ vertices of a graph G.

- Minimum spanning tree < Prim's, Kruskal's
- Dijkstra's algorithm
- Worshall's algorithm

(1) Minimum Spanning Tree –

A spanning tree of a connected, undirected graph G is a sub-graph of G which is a tree that connects all the vertices together. A graph can have multiple spanning tree.

A minimum spanning tree, is consider in a weighted graph, where a spanning tree with minimum weight.

OR A MST is a spanning tree that has weight associated with its edge and the total weight of the tree (the sum of edges weight) is minimum.

Def$^n$

Spanning tree - In an undirected and connected graph $G = (V, E)$, a spanning tree is a tree that spans (means it include every vertex of $G$) and is a subgraph of $G$.

Minimum spanning tree

The cost of the spanning tree is the sum of the weight of all the edges in the tree. There can be many spanning tree. Minimum spanning tree where the cost is minimum among all the spanning tree. There may be many minimum spanning tree.

Application are -

Cluster Analysis
Handwriting recognition
Image segmentation

Two Algorithm - for finding the mst

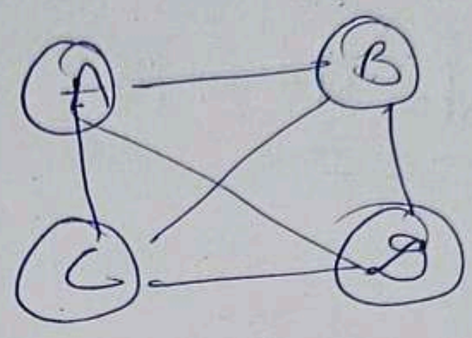
(1) Prim's $O((V+E) \log V)$
(2) Kruskal $O(E \log V)$

Swami Keshvanand Institute of Technology, Management & Gramothan,
Ramnagaria, Jagatpura, Jaipur-302017, INDIA
Approved by AICTE, Ministry of HRD, Government of India
Recognized by UGC under Section 2(f) of the UGC Act, 1956
Tel. : +91-0141- 5160400 Fax: +91-0141-2759555
E-mail: info@skit.ac.in Web: www.skit.ac.in

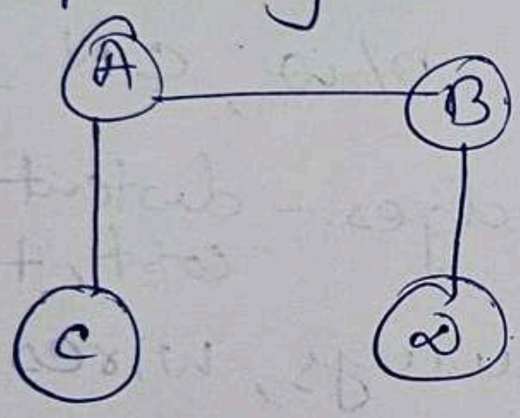
असतो मा सद्गमय

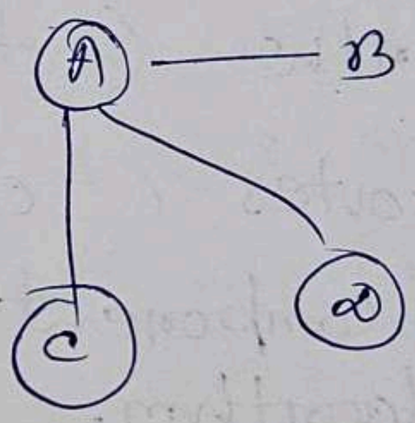
* A spanning tree, is a acyclic graph but connected with all vertices

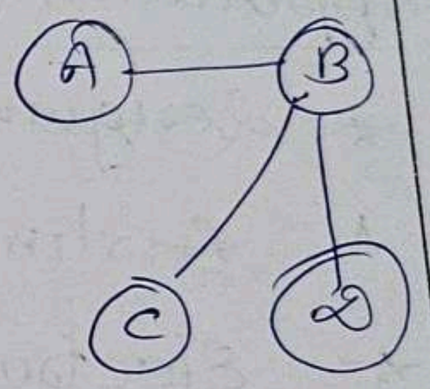
Graph -

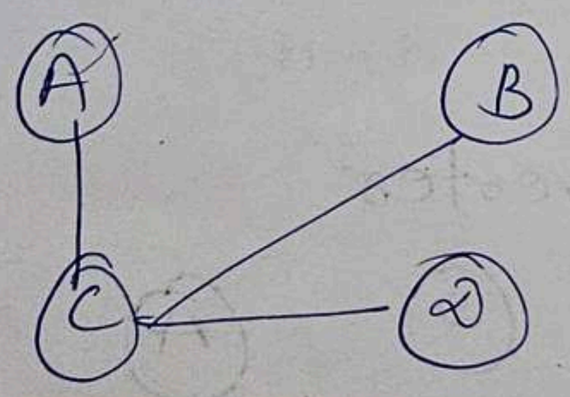


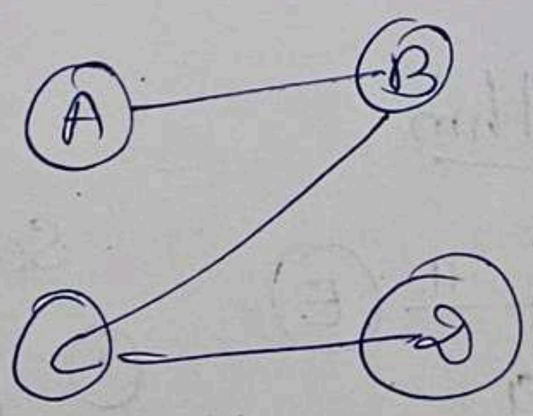

spanning trees are

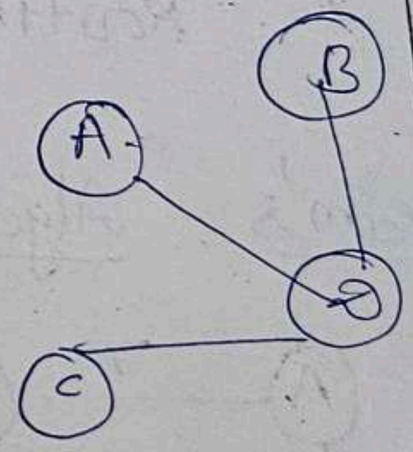


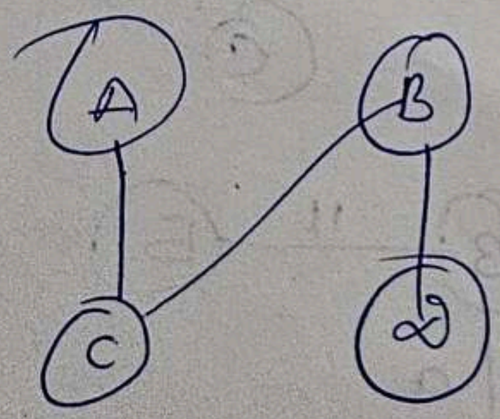



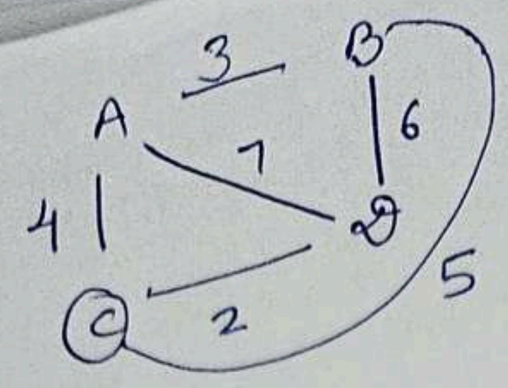



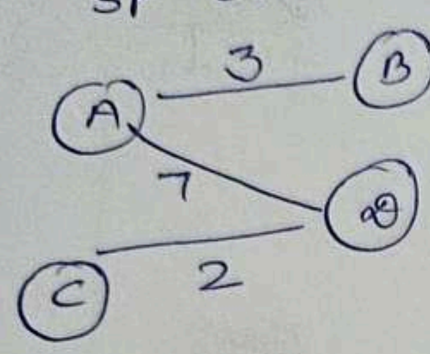



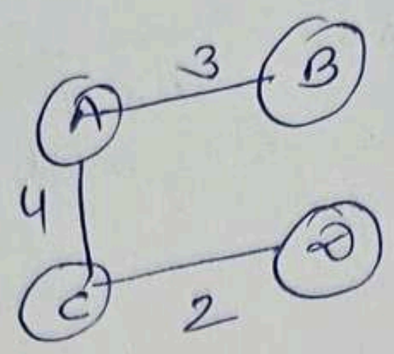
MST - Graph



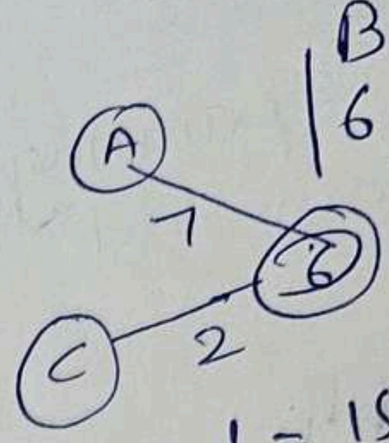

SP are



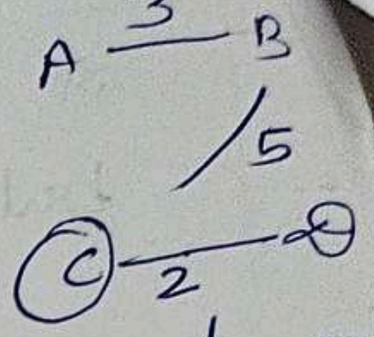

Cost = 12




Cost = 9




Cost = 15

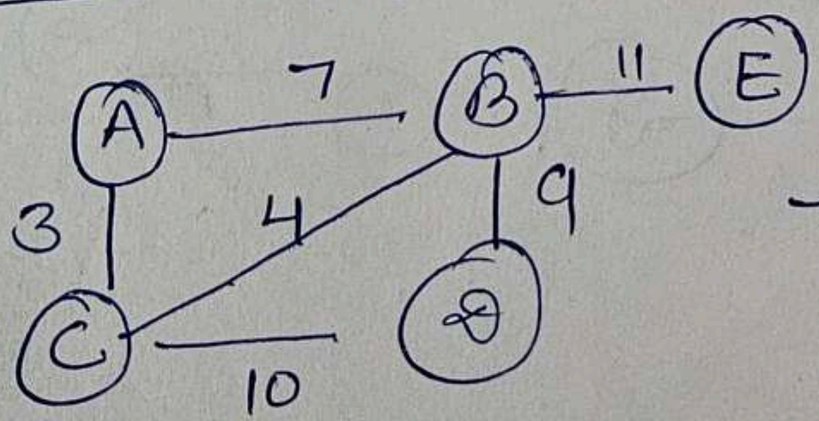



Cost = 10

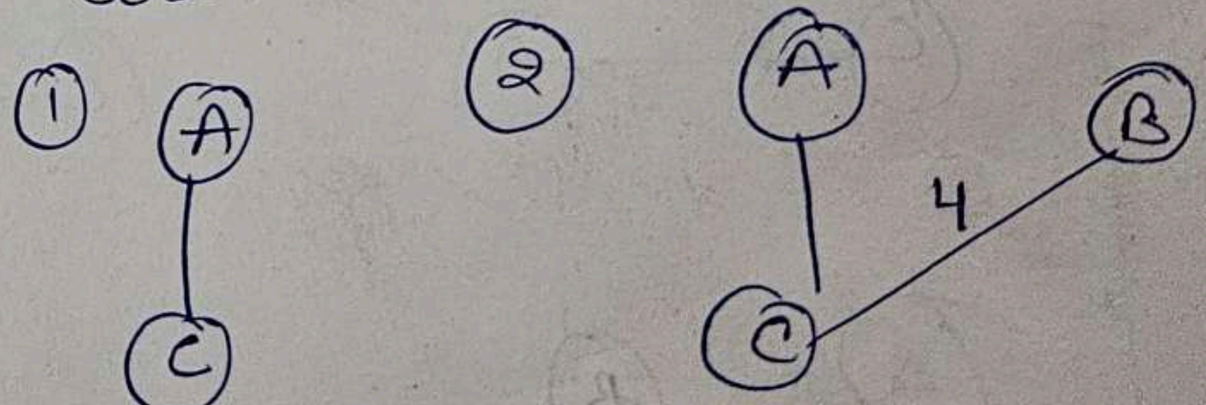
Application

* Designing n/w ( telephone n/w, CN )
* Airline routes ( cities, edges - distant, cost, time)
* Electronic component, railways, wires
* Routing algorithm.

Prim's Algorithm




Source vertex




Cost = 27

Swami Keshvanand Institute of Technology, Management & Gramothan,
Ramnagaria, Jagatpura, Jaipur-302017, INDIA
Approved by AICTE, Ministry of HRD, Government of India
Recognized by UGC under Section 2(f) of the UGC Act, 1956
Tel. : +91-0141- 5160400 Fax: +91-0141-2759555
E-mail: info@skit.ac.in Web: www.skit.ac.in

असतो मा सद्गमय

PRIM - MST ( V, E, w, s)

```
1.  foreach v ∈ V
2.      do key[v] ← ∞
3.         P[v] ← NIL
4.  key[s] ← 0
5   Q ← V
6   while Q ≠ ϕ
7       do u ← Extracted-Min(Q)
8          foreach v ∈ Adj[u]
9              do if v ∈ Q and w(u,v) < key[v]
10                then key[v] ← w(u,v)
11                     P[v] ← u
12. return P
```

| step 1 to 3 | A | B | C | D | E |
|---|---|---|---|---|---|
| key | ∞ | ∞ | ∞ | ∞ | ∞ |
| | NIL | NIL | NIL | NIL | NIL |

④

| A | B | C | D | E |
|---|---|---|---|---|
| 0 | ∞ | ∞ | ∞ | ∞ |

⑤ Q ← V  Q: A, B, C, D, E
0 ∞ ∞ ∞ ∞

⑥ u ← A

| for | A | B | C | D | E |
|---|---|---|---|---|---|
| | 0 | 7 | 3 | ∞ | ∞ |
| P | | A | A | | |

Q — B, D, E
C ← C  7 | 3 | ∞ | ∞
C — D, B, A

| | A | B | D | E |
|---|---|---|---|---|
| | 0 | 4 | 10 | ∞ |

U ← B

MST - ^ 3 -

MST_ kruskal ( G, w )

1. A = ϕ
2 for each vertex v ∈ G. V
3 make_ set (v)
4 Sort the edges of E(G) into non-decreasing order by weight w
5 for each edge (u,v) ∈ E(G) taken in non-decreasing order by weight
6 if find- set(u) ≠ find set (v)
7 A = A ∪ {(u,v)}
8 Union (u,v)
9 return A.

38, 39, 35, 37, 40,
47, 46, 50, 49, 43
55, 60, 62, 63, 65, 64

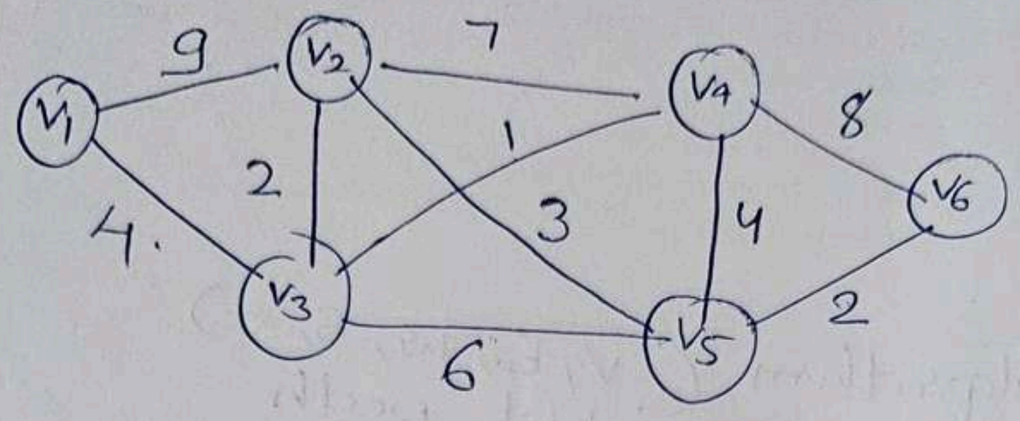
1, 12
8, 4, 5, 10, 11,
15, 13, 16, 18, 21
24, 25, 26, 27, 26,

Swami Keshvanand Institute of Technology, Management & Gramothan,
Ramnagaria, Jagatpura, Jaipur-302017, INDIA
Approved by AICTE, Ministry of HRD, Government of India
Recognized by UGC under Section 2(f) of the UGC Act, 1956
Tel. : +91-0141- 5160400 Fax: +91-0141-2759555
E-mail: info@skit.ac.in Web: www.skit.ac.in

Dijkstra's Algorithm (V, E, w, S, Q)
// Single source shortest path

1. for each v ∈ V
2. do dist[v] = ∞ (distance)
3. P P[v] = NIL (-1)
4. dist(S) = 0
5. PQ is a Priority Queue
6. PQ ← V (All nodes with its dist value)
7. while (PQ is not empty) do
8. u ← Extracted- min(PQ)
9. for each neighbor v of u do
10. w = weight of edge(u,v)
11. new dist = dist[u] + w
12. if (new dist < dist[v]) then
13. dist[v] = dist[u]+w
14. P[v] = u
15. Stop.

Complexity — O(|E| log V) / O(V²)
Priority Queue — Array




S - $v_1$
D - $v_6$

① 

| | $v_1$ | $v_2$ | $v_3$ | $v_4$ | $v_5$ | $v_6$ |
|---|---|---|---|---|---|---|
| dis | 0 | ∞ | ∞ | ∞ | ∞ | ∞ |
| P | NIL | -1 | -1 | -1 | -1 | -1 |

② 

| $v_1$ | $v_2$ | $v_3$ | $v_4$ | $v_5$ | $v_6$ |
|---|---|---|---|---|---|
| | 9 | 4 | ∞ | ∞ | ∞ |
| | $v_1$ | $v_1$ | -1 | -1 | -1 |

③ 

| $v_3$ | $v_2$ | $v_4$ | $v_5$ | $v_6$ |
|---|---|---|---|---|
| | 6 | 5 | 10 | ∞ |
| | $v_3$ | $v_3$ | $v_3$ | -1 |

④ 

| $v_4$ | $v_2$ | $v_5$ | $v_6$ |
|---|---|---|---|
| | 6 | 9 | 13 |
| | $v_3$ | $v_4$ | $v_4$ |

⑤ 

| $v_2$ | $v_5$ | $v_6$ |
|---|---|---|
| | 9 | 13 |
| | $v_4$ | $v_4$ |

⑥ 

| $v_5$ | $v_6$ |
|---|---|
| | 11 |
| | $v_5$ |

⑦ $v_6$ — Empty

B

$v_1$ - $v_3$ - $v_4$ - $v_5$ - $v_6$

Total cost (11)

=27